All Nations Theological College & Seminary

ANTS

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Unit 2: Computer Systems

COMPUTER LITERACY

What is a computer?

- A computer is an **electronic device** which inputs data, processes data, and outputs information when given a set of instructions. Computer instructions are known as **commands.**
- Computers are controlled by programmed instructions that transform the data into meaningful information.
- Generally, a Computer is a **device** that accepts **input**, **processes** it, **stores** data, and produces **output**.

Parts of a computer

Characteristics of a computer

Computers have several key characteristics that define their functionality and utility, which include the following;

- **Speed:** computers are very fast in their operations and performs tasks in microseconds or faster.
- **Accuracy:** Computers are known to be very accurate and hardly make mistakes. The error are usually due to human error or inaccurate data entered. If wrong data is entered, wrong results are obtained hence the saying Garbage In Garbage out (GIGO).
- **Storage Capacity:** For a computer to be able to work, it must have some form of work space where data is stored before being processed or information is stored before being output to particular devices. The storage is called Memory. **Automation:** Performs operations automatically once programmed.
- **Versatility:** Computers have the capacity of performing different kinds of work depending on the input. A computer can be used for instance to process images captured, generate reports and prepare payroll.
- **Diligence (Consistency): :** Computers have the ability to perform the same task over and over again for a long time without getting tired.
- **Scalability:** Easily upgraded or expanded to meet growing needs.
- **Multitasking:** Can perform multiple tasks simultaneously.
- **Artificial Intelligence :** Computers are artificially intelligent. They can respond to requests given and then provide solutions quickly.

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Advantages of using Computers

- ❖Computers with communicating capability can share data and information with other computers.
- ❖Tasks can be completed faster because computers work at an amazing speed.
- ❖Computers can store enormous amounts of data for various purposes.
- ❖The high reliability of components inside the modern computers enables computers to produce consistent results.
- ❖Tasks can be completed with little human intervention. i.e. automatic.
- ❖Overall security and accuracy can be raised due to less human intervention.
- ❖Reduced costs. In an automated office environment, a computer is used to perform tasks that would normally be assigned to several officers.
- ❖Offers convenient and paperless storage within it this saves companies a lot of floor space that would otherwise be sed to store files.

Disadvantages of using Computers

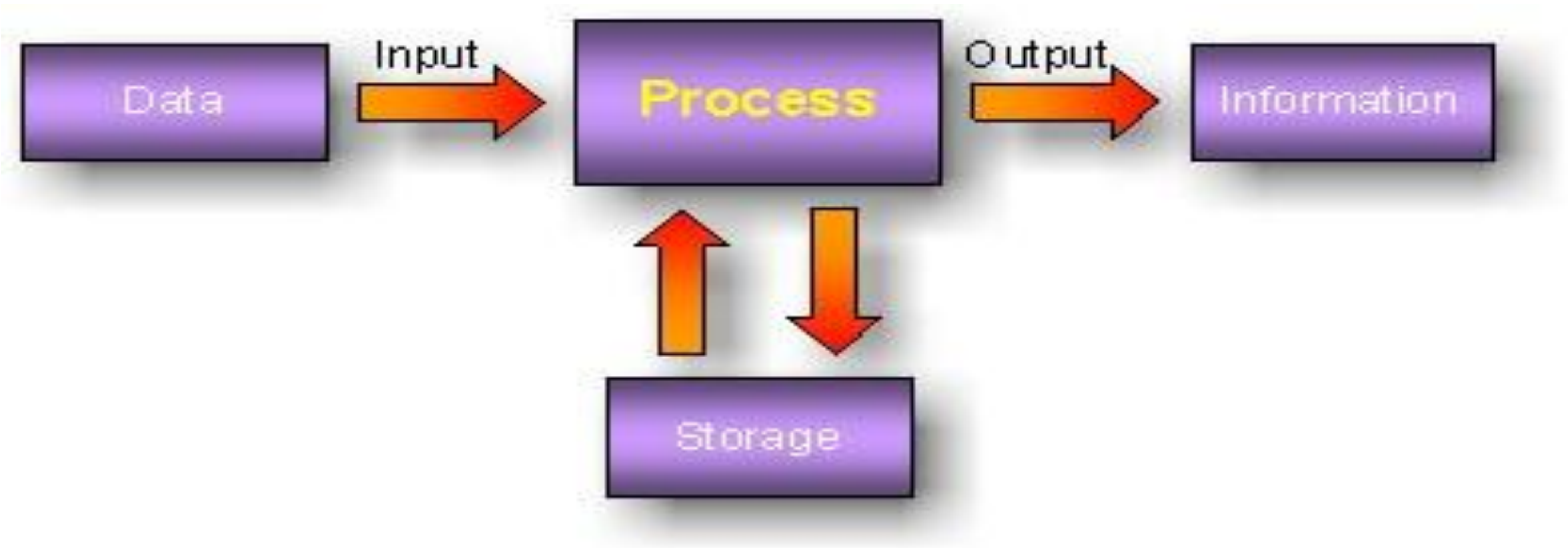
- Computers are expensive to acquire.
- Extra cost is required to employ specialized staff to operate and manage Computers.
- By doing enormous work in a short time, Computers can lead to unemployment.
- Training and re-training of users on how to use new technologies can be expensive.
- Easy transmission of viruses via the internet, which may lead to creating untimely costs to the recipient and sender computers.
- Problems may arise when computers cannot be used either because they are malfunctioning or damaged. This can bring an organization to a halt if no back up exists.
- Face to face interaction among people may be reduced.
- Computers and automation can lead to loss of jobs especially for computer illiterate people

The four main functions of a computer

- The four main functions of a computer, often referred to as the **information processing cycle are;**

1. **Input;** involves receiving data and instructions from various input devices or sources such as keyboard, mouse, etc.
2. **Processing;** the function where the computer's central processing unit (CPU) takes the input data and performs operations on it according to the instructions provided.;
3. **Storage;** saving data and instructions for immediate or future use.
4. **Output;** the function where the processed data is presented to the user or sent to another system.

The information processing cycle



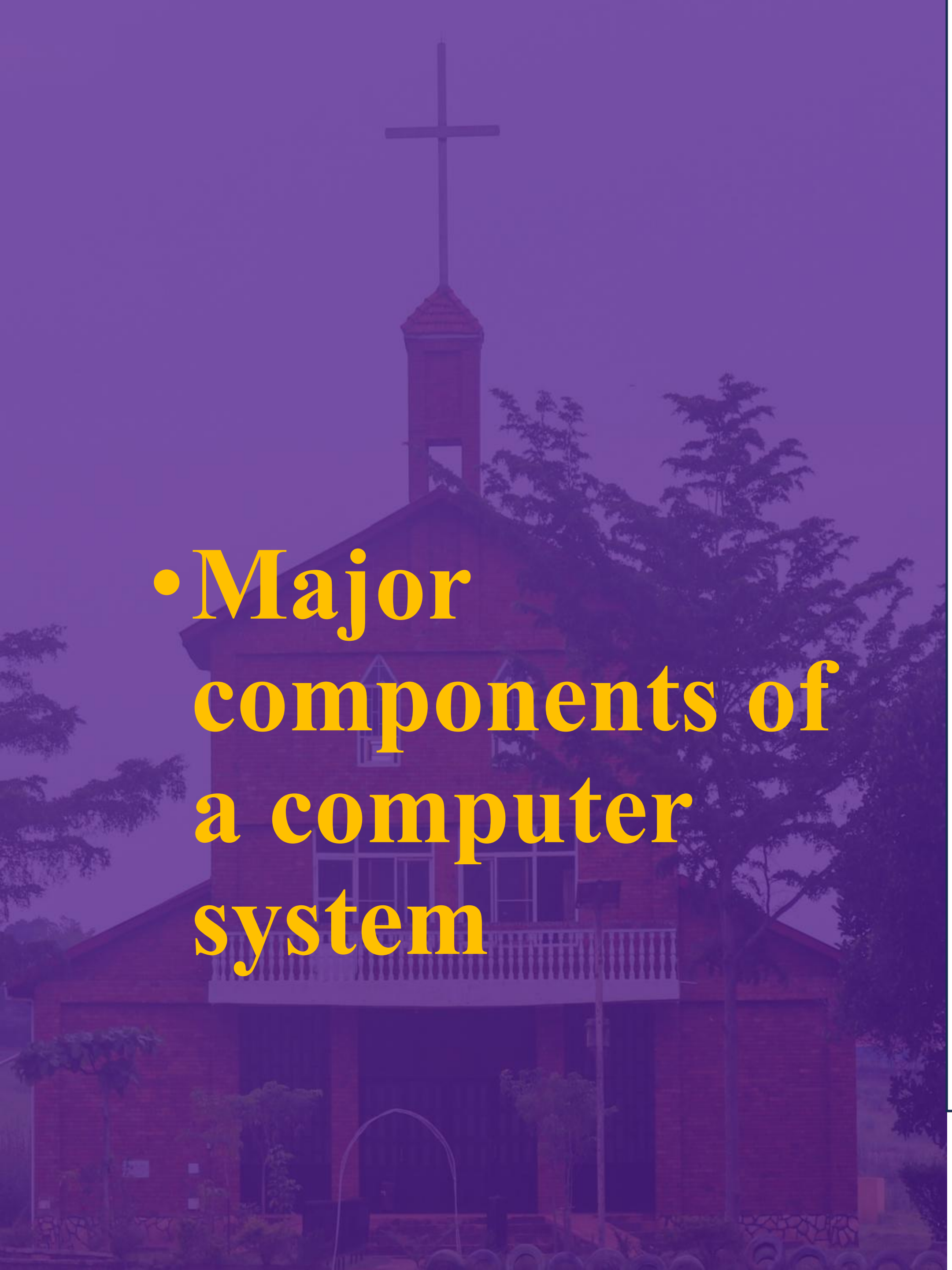
A Computer System

- A system is made up of different components, all working in harmony to achieve a certain objective. Some examples of systems include the digestive system, the respiratory system, and the educational system among others. Systems are made up of smaller systems known as subsystems. The computer is also a system.
- A computer system consists of a set of devices that are designed to receive and process data. The result is information that is generated, and is known as the **output**. The computer system consists of three major components namely the **hardware**, the **software** and the **liveware**.

Functional Organization of a Computer System

- The computer system is organized into 3 main components: the computer hardware, the computer software, and the liveware/user.

1. **Hardware:** The hardware component includes the tangible parts such as the input devices, the output devices and the storage devices. Hardware, therefore, consists of the physical and tangible components that make up the computer system.
2. **Software:** Software refers to the intangible components, which are also called programs or applications.
3. **Liveware(User):** This refers to the computer user, who is also known as orgware or the human ware. The user commands the computer system to execute an instruction, for example, copy, print, send, and calculate among others.

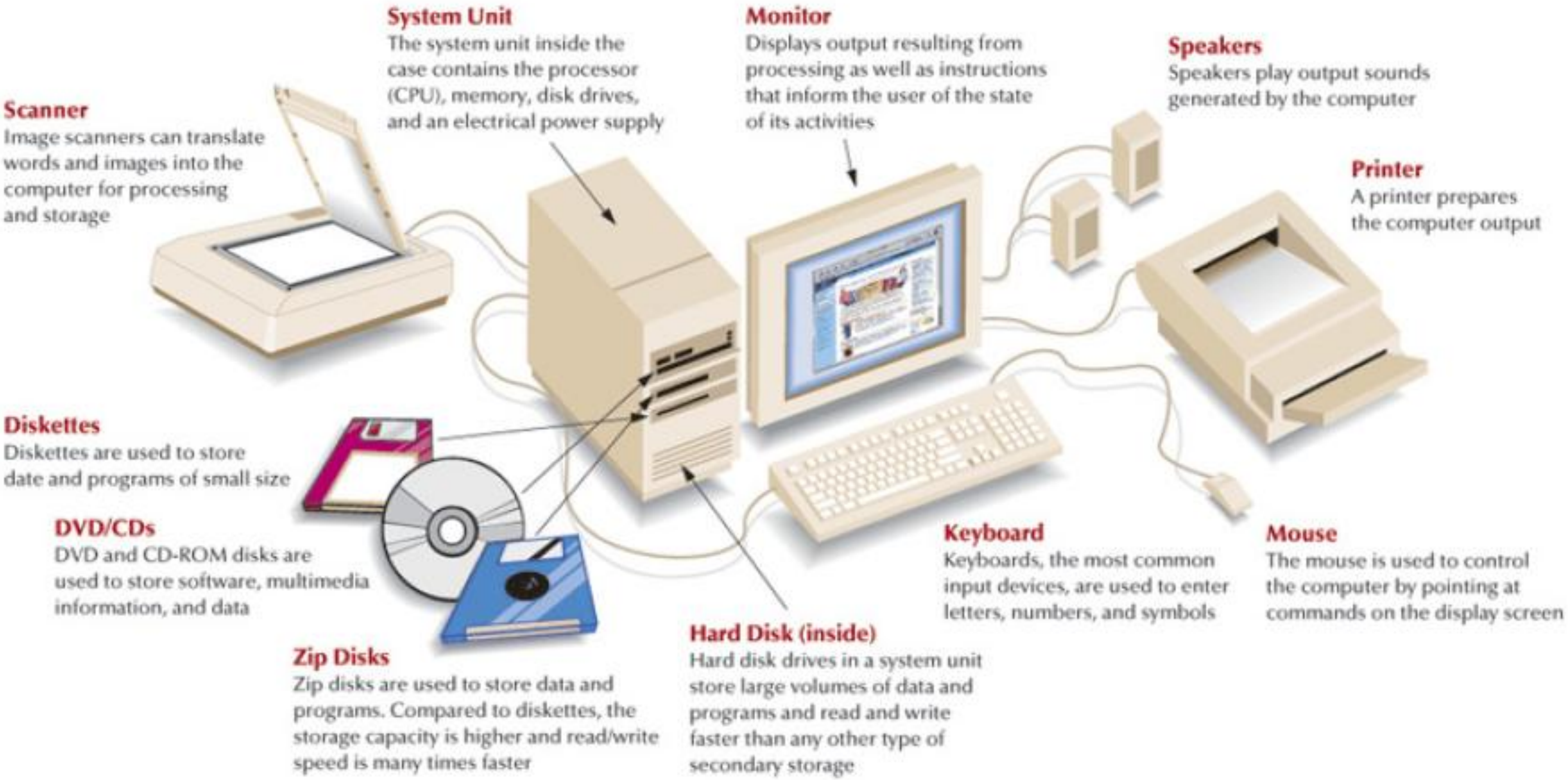


• Major components of a computer system

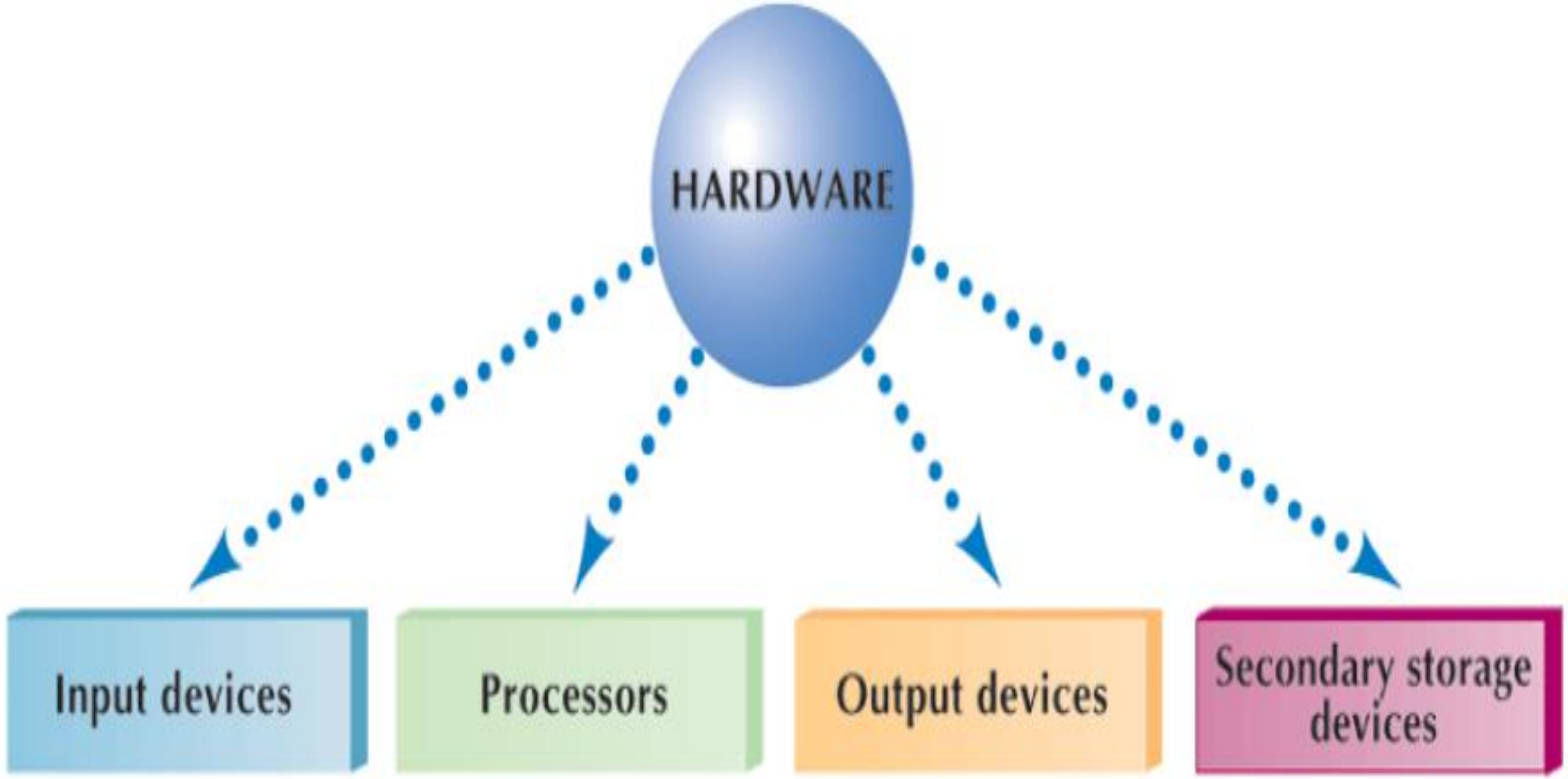
1. Hardware

- A computer's hardware consists of electronic devices; the parts you can see and touch. In other words, these are tangible parts of a computer.
- The term "device" refers to any piece of hardware used by the computer, such as a keyboard, monitor, modem, mouse, etc.
- Computer hardware is classified into 4 categories, **input, output, processing and storage devices.**

Examples of Hardware



Categories of hardware



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Take Home 1!!

Give at least 3 examples for each of the following categories of hardware devices

- Input devices
- Output devices
- Storage devices
- Processor

Memory/storage devices of a computer

Memory devices are used for storing data and information in a computer.

Types of Memory Devices

1. Primary Memory (Main Memory):

- **RAM (Random Access Memory):** Temporary and volatile, used for running programs.
- **ROM (Read Only Memory):** Permanent memory; stores startup instructions (firmware).

2. Secondary Memory (Storage):

- Non-volatile and long-term storage.
- Examples: Hard Disk Drives (HDD), Solid State Drives (SSD), Optical Discs (CD/DVD), Flash disks etc.

RAM vs ROM

- Random Access Memory
- Volatile - data lost when power off
- Can read and write data continuously
- Temporary workspace for active programs
- Running applications, OS components, user data

- Read-Only Memory
- Non-volatile - data retained without power
- Data is permanent, difficult to modify
- Stores permanent system instructions
- BIOS/UEFI, firmware, startup instructions

- **Major components of a computer system**

2. Computer software

- **Software** is a set of instructions that drive a computer to perform specific tasks. Unlike hardware software is intangible. It can't be seen and touched.
- These instructions tell the machine’s physical components what to do.
- A set of instructions is often called a **program**.
- When a computer is using a particular program, it is
- said to be **running** or **executing** the program.

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Continuation…

Computer software can be broadly classified into two categories i.e.

- Systems software
- Application software

1. **System software**: Is any program that controls the computer's hardware or that can be used to maintain the computer in some way so that it runs more efficiently.

System software include;

An **operating system; this** tells the computer how to use its own components. All computers require an operating system to operate.

The popular types of operating systems are:

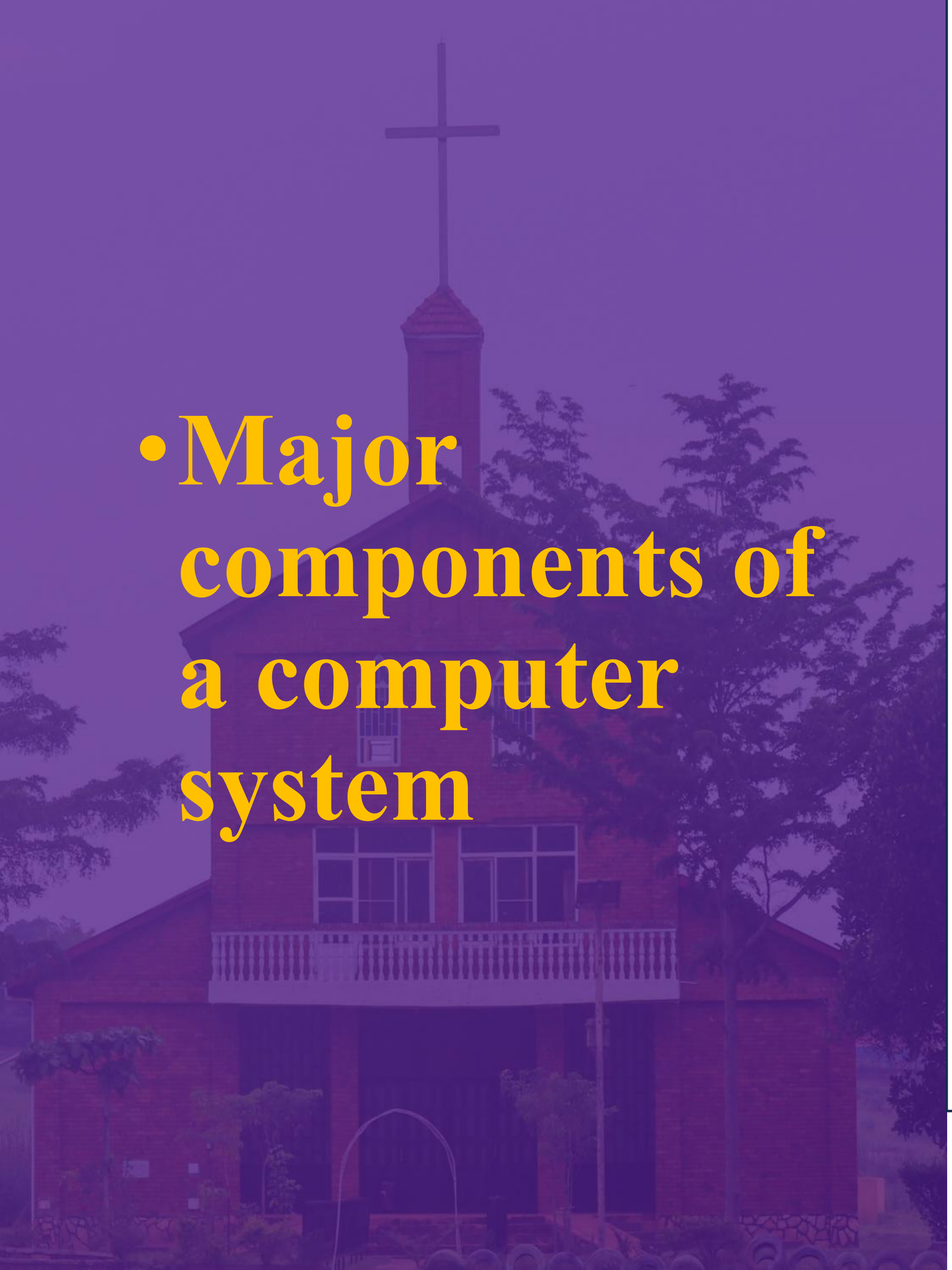
- Windows operating system
- Macintosh operating system (Mac OS)
- Linux.

Continuation…

2. Application Software

These are programs designed to meet user specific needs. Example include presentation software such as

- PowerPoint
- word processors such as MS Word
- spreadsheets such as Excel, etc.



- **Major components of a computer system**

3. Liveware: Users and Creators of IT Applications

- **User (End User):** The people who use computers in their jobs or personal lives
- **Programmer/Analyst:** A person who has joint responsibility for determining system requirements and developing and implementing the systems.
- **Computer Engineer:** Professional who designs, develops, and oversees the manufacturing of computer equipment.
- **Systems Engineer:** Professional who installs and maintains hardware.

Other components of a computer system

- **Data** is raw, unorganized facts or figures.
- Data can be numbers, characters, symbols, or raw observations.
- Data has no meaning by itself until it is processed.
- **Information** is processed and organized data that has meaning.
- Information is presented as statements, summaries, or conclusions.
- Information is data interpreted within a specific context, it is considered meaningful and useful.
- Information is useful for making decisions and solving problems.

Data and information can be stored in a computer in **files**

Data vs Information

- While the terms are used interchangeably, they represent different stages of the processing cycle.
- **Data**(Input) → Processing (Sorting, calculating, editing, organizing) → **Information** (Output)

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Data vs Information

- **Data**

- The raw material
- No inherent meaning on its own
- Raw facts, figures and symbols
- Used as an input for processing
- Example, a list of random test scores

- **Information**

- The finished product
- Meaningful and easy to understand
- Organized, structured and processed
- Used for decision making and action
- Example, a report showing the class average.

FILES

A **file** is a named collection of data, stored on a storage medium such as a hard disk.

There are two types of **files**

- **Data files** contains text, images, or other data that can be used by a program.
- **Executable files** contains programs or instructions that tell the computer how to perform a task.

File Extension

- A **file extension** is a set of three or four letters after the dot in a filename. The file extension indicates what kind of file it is. For example, **myfile.txt** is a text file probably created using notepad.

Purpose:

- Helps the operating system know which software to use to open the file.
- Identifies the format and type of the file (e.g., document, image, audio, video, etc.)

Examples of Common File Extensions

Extension	File Type
.doc	Word document
.docx	Word Document (modern)
.pdf	Portable Document Format
.txt	Text File
.png	Image File (lossless)
.mp3	Audio File
.mp4	Video File

FOLDER

- A **folder** is a **virtual container** used to **store and organize files** and sometimes other folders on a computer or storage device.
- Folders stored within other folders are known as subfolders

Why Use Folders?

- Better organization (e.g., keeping work, school, and personal files separate)
- Faster file access
- Simplifies backup and transfer of data
- Improves productivity and reduces file clutter

Each folder can contain multiple files or more folders, making file management easier and cleaner.

Creating a folder

1. Go to the location where you want to create your folder.
2. Eg to create a folder on the desktop, right click on the desktop, select new followed by folder, a new folder appears with a default name New Folder, give it a name of your choice and press enter
3. You can open the new folder by double clicking its icon.

OR You can use a shortcut

1. Go to the folder or location where you wish to create a new folder
2. Press Ctrl + Shift + N (Press each followed by the other in that order)
3. Rename the new folder created

THE END

THANK YOU!!